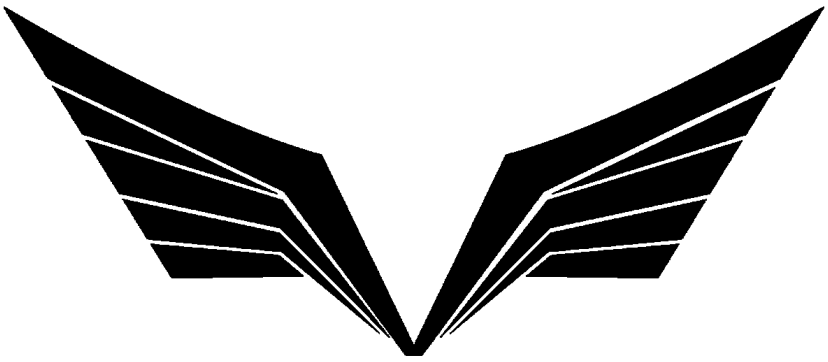


SUN DEVIL ROCKETRY

Avionics Development Team

New Liquids Feed System (NLFS)
System Requirements Document

Project:	New Liquids Architecture
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1. OVERVIEW:

The previous liquid avionics platform required two expensive, custom PCBs as well as all of the hardware and software work needed to build and test them. Every change to a sensor required a full rebuild of the hardware, and every rebuild of the hardware required firmware to be brought back up and tested. There were also significant reliability and maintainability issues on this platform.

So, at the conclusion of the 2025-26 development cycle, the Avionics group at Sun Devil Rocketry has set out to replace this aging system. In particular, the group is looking for a solution to the following pitfalls of last generation:

1. Long hardware lead times
2. Monolithic hardware + firmware components
3. Slow data logging performance
4. Unreliable hardware connections

The solution the group has come up with is a distributed system with a consumer off the shelf (COTS) data acquisition and control device at its core.

2. HARDWARE REQUIREMENTS

2.1: <*POSTPONED*>The system shall be designed to support a liquid engine powered rocket.

2.1.1: <*POSTPONED*>The system shall use the MAV and its extensions when data must be collected onboard the rocket.

2.1.2: The system shall use COTS products for on-ground data acquisition and control.

2.2: The system shall integrate the LabJack T7.

2.2.1: The LabJack T7 shall be used to collect data.

2.2.2: The LabJack T7 shall be used to provide control inputs.

2.3: The system shall have its hardware inputs and outputs labeled at the connector.

Note: As part of the hotfire procedure, a wiring diagram for the system's specific configuration should be produced with all inputs and outputs to the avionics specified.

2.4: The system shall have a hardware mediated killswitch that does not require software input.

2.4.1: The system shall be able to shut off when a mechanical failure occurs.

2.5: <*POSTPONED*>The system shall produce multiple consecutive sources of telemetry data.

2.5.1: <*POSTPONED*>The system shall feed onboard data to the flight computer via CAN.

2.5.2: The system shall be able to collect readings from pressure transducers through the LabJack.

2.5.3: The system shall be able to collect readings from thermocouples through the LabJack.

2.5.4: The system shall be able to log data at a minimum rate of 200 Hz.

2.6: <*POSTPONED*>The system shall have redundant methods of data storage for data collected in flight.

2.6.1: <*POSTPONED*>An external flash chip shall be used to store data onboard.

2.6.2: <*POSTPONED*>Data shall be transmitted through LoRa.

2.7: The system shall have connections for every component that requires one.

2.7.1: <*POSTPONED*>The system shall use "quick disconnects" for check valves during launch.

2.7.2: All sensor and actuator wiring shall terminate at screw-clamp terminal blocks rated for a minimum of 25A / 300V, accepting 12-22 AWG wire.

2.7.3: All cables entries into enclosures (LabJack T7, relay board holders, expansion cards) shall incorporate strain-relief cord grips or cable glands, verified by visual inspect during system integration

2.7.4: <*POSTPONED*>The LabJack T7 enclosure and relay board enclosures shall meet a minimum IP65 rating for dust and water resistance

2.8: The system's electrical design shall meet the following signal integrity and protection requirements:

- 2.8.1: All thermocouple channels shall use external signal conditioning (e.g., LJTick-InAmp, AD8495, or MAX31856) to buffer & amplify the signal to level compatible with LabJack's configured input range.
- 2.8.2: Every solenoid actuator driver circuit shall include flyback (TVS or diode) suppression across the coil to prevent inductive transients from damaging the LabJack or relay boards.
- 2.8.3a: All electronics enclosures and equipment chassis shall be bonded to a common earth ground reference point, with bonding resistance verified to be less than 10Ω. *Verification Plan: Continuity test each chassis and system designed earth ground point.*
- 2.8.3b: All analog sensor cabling shall use shields twisted-pair (STP) cable, with the shield terminated to chassis at one end only.
- 2.8.4: <*POSTPONED*>Control outputs for solenoid actuators should be galvanically or optically isolated from the DAQ analog measurement ground where practical.

3. SOFTWARE REQUIREMENTS:

3.1. The system shall be written in a high-level programming language.

3.1.1. The chosen language shall support an “exception” or “try/catch” construct for graceful error handling.

3.2. The system shall feature a graphical user interface as the primary mode of operation.

3.2.1. The interface shall display a simplified P&ID containing all inputs (sensor readings) and outputs (control commands) available to the system.

3.2.2. The interface shall use color to convey information about the system status in accordance with the table below:

Reading/Control	Color	Meaning
Control	Green	The control interface is open.
	Red	The control interface is closed.
Reading	Green	The sensor reading is within the “normal” range, or the connection is open.
	Orange	The sensor reading is within the “warning” range, or the connection is unstable.
	Red	The sensor reading is outside both the “normal” and “warning” ranges, or the connection has disconnected.

Table 1: Global Interface Color-Coding Scheme

3.2.2.1: The colors specified in the Global Interface Color-Coding Scheme shall ONLY be used in the interface for the specified purposes.

3.2.3. The interface shall provide a large “**ABORT**” option that sends control inputs to immediately return the system to a safe state.

3.2.3.1. If there are “warning” level system messages, the “**ABORT**” option shall be highlighted in orange.

3.2.3.2. Otherwise, if the system is not in a safe state, the “**ABORT**” option shall be highlighted in yellow.

3.2.4. The interface shall provide some immediate feedback to all control inputs, even if processing of the command is delayed.

3.2.4.1. If processing of a control input is delayed, the system shall provide feedback when processing of that input is completed.

3.2.5. The interface shall provide elements on the main page to manage external connections.

3.2.5.1. The interface shall provide an element on the main page to manage the LQD-DAQ connection.

3.2.5.2. <*POSTPONED*>The interface shall provide an element on the main page to manage the SDEC-API connection.

3.2.6. The interface shall display the current status of the external connections.

3.2.7. The interface shall display a rolling chart for the following readings:

3.2.7.1. Chamber pressure

3.2.7.2. Thrust

3.2.7.3. LOx Tank Pressure

3.2.8. The interface shall display all of the following derived readings:

3.2.8.1. Mass flow rate (both propellants)

3.2.8.2. Mixture ratio O/F

3.2.8.3. Total impulse (load cells OR estimate for flight capable deployments)

3.2.9. <*POSTPONED*>The interface shall display an alert for each of the following conditions:

3.2.9.1. <*POSTPONED*>Off-nominal readings (outside warning and normal ranges)

3.2.9.2. <*POSTPONED*>Tank pressure below LOX saturation

3.3. <*POSTPONED*>The system shall support a method to interface with SDEC-API.

3.3.1. <*POSTPONED*>The system shall support simultaneous operations with other API clients.

Note: See SDEC-API#15

3.3.2. <*POSTPONED*>The system shall push new data to connected UI clients at 60 hz or less.

3.3.3. <*POSTPONED*>The system shall support operation without SDEC-API connected.

3.3.4. <*POSTPONED*>The system shall handle unexpected SDEC-API disconnects without impacting the rest of the system.

3.4. The system shall be able to log data from the LabJack.

3.4.1. The logging system shall be uncoupled from the interface (e.g. support a higher logging rate than what is displayed)

3.5. The system shall support a debug console for logging of events in the system.

3.6. <*POSTPONED*>The system must satisfy the following ease of use requirements:

3.6.1. <*POSTPONED*>The UI shall be deployable on each client with one click.

3.6.2. <*POSTPONED*>The UI shall not require any installation after downloading.

3.6.3. <*POSTPONED*>The backend shall be deployable with one command.

3.6.4. <*POSTPONED*>The backend shall be installable with one command.

3.6.5. The interface shall have a settings page to change the following features:

3.6.5.1. Display Units

3.6.5.2. Light/Dark mode

3.6.5.3. Aural Alert Enablement

3.6.6. The interface's user configurable settings shall be restored between runs.

4. REVISION HISTORY:

1.0 - Initial Revision